

Data Base Lab
Lab Task 10



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Section 4A

Question 01

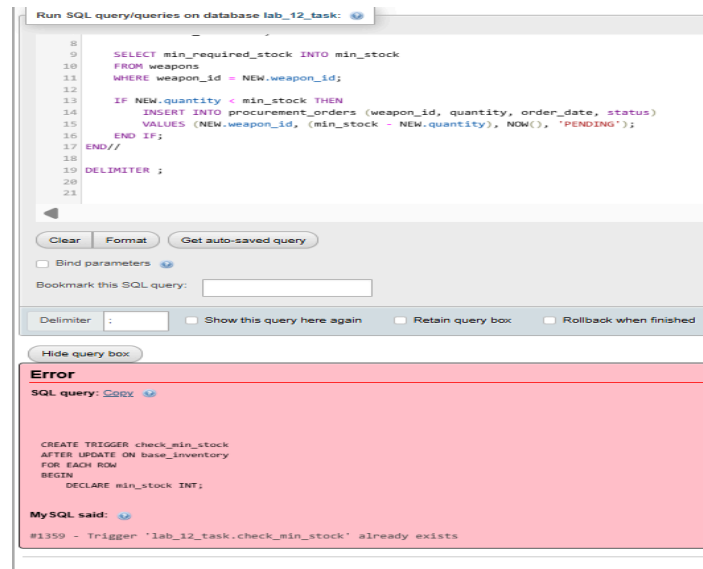
DELIMITER //

```
CREATE TRIGGER check_min_stock
AFTER UPDATE ON base_inventory
FOR EACH ROW
BEGIN
    DECLARE min_stock INT;

    SELECT min_required_stock INTO min_stock
    FROM weapons
    WHERE weapon_id = NEW.weapon_id;

    IF NEW.quantity < min_stock THEN
        INSERT INTO procurement_orders (weapon_id, quantity, order_date, status)
        VALUES (NEW.weapon_id, (min_stock - NEW.quantity), NOW(), 'PENDING');
    END IF;
END//
```

DELIMITER ;



```
UPDATE base_inventory SET quantity = 70 WHERE base_id = 1
AND weapon_id = 1;
SELECT * FROM procurement_orders;
```

Showing rows 0 - 3 (4 total. Query took 0.0025 seconds.)

SELECT * FROM base_inventory;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

 | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

base_id

weapon_id

quantity

Edit

Copy

Delete

1150

Edit

Copy

Delete

1220

Edit

Copy

Delete

2140

Edit

Copy

Delete

235

Check all

 | With selected:

Edit

Copy

Delete

Export

Show all

 | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print

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Display chart

Create view

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1 row affected. (Query took 0.0004 seconds.)

UPDATE base_inventory SET quantity = 70 WHERE base_id = 1 AND weapon_id = 1;

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 0 (1 total. Query took 0.0006 seconds.)

SELECT * FROM procurement_orders;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

 | Number of rows: 25 | Filter rows: Search this table

1 row affected. (Query took 0.0004 seconds.)

UPDATE base_inventory SET quantity = 70 WHERE base_id = 1 AND weapon_id = 1;

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 0 (1 total. Query took 0.0006 seconds.)

SELECT * FROM procurement_orders;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

 | Number of rows: 25 | Filter rows: Search this table

Extra options

order_id

weapon_id

quantity

order_date

status

Edit

Copy

Delete

1110 2026-04-27 18:39:51 PENDING

Check all

 | With selected:

Edit

Copy

Delete

Export

Show all

 | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

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Console

Question 02

DELIMITER //

```
CREATE TRIGGER process_transfer
AFTER INSERT ON transfers
FOR EACH ROW
BEGIN
    UPDATE base_inventory
    SET quantity = quantity - NEW.quantity
    WHERE base_id = NEW.from_base AND weapon_id = NEW.weapon_id;

    UPDATE base_inventory
    SET quantity = quantity + NEW.quantity
    WHERE base_id = NEW.to_base AND weapon_id = NEW.weapon_id;
END//

DELIMITER ;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0076 seconds.)

```
CREATE TRIGGER process_transfer AFTER INSERT ON transfers FOR EACH ROW BEGIN -- Deduct from source base UPDATE base_inventory SET quantity
= quantity - NEW.quantity WHERE base_id = NEW.from_base AND weapon_id = NEW.weapon_id; -- Add to destination base UPDATE base_inventory
SET quantity = quantity + NEW.quantity WHERE base_id = NEW.to_base AND weapon_id = NEW.weapon_id; END;
```

[Edit inline] [Edit] [Create PHP code]

```
INSERT INTO transfers (weapon_id, from_base, to_base, quantity, transfer_date)
VALUES (1, 1, 2, 10, NOW());
SELECT * FROM base_inventory;
```

✓ 1 row inserted.

Inserted row id: 1 (Query took 0.0003 seconds.)

```
-- Insert a transfer INSERT INTO transfers (weapon_id, from_base, to_base, quantity, transfer_date) VALUES (1, 1, 2, 10, NOW());
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 3 (4 total, Query took 0.0002 seconds.)

```
SELECT * FROM base_inventory;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				base_id	weapon_id	quantity
☐	 Edit	 Copy	 Delete	1	1	60
☐	 Edit	 Copy	 Delete	1	2	20
☐	 Edit	 Copy	 Delete	2	1	50
☐	 Edit	 Copy	 Delete	2	3	5

↑ ☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

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Question 03

DELIMITER //

```
CREATE PROCEDURE redistribute_stock(
    IN weapon_id_input INT,
    IN quantity_input INT
)
BEGIN
    DECLARE alpha_stock INT;
    DECLARE bravo_stock INT;

    SELECT quantity INTO alpha_stock
    FROM base_inventory
    WHERE base_id = 1 AND weapon_id = weapon_id_input;

    SELECT quantity INTO bravo_stock
    FROM base_inventory
    WHERE base_id = 2 AND weapon_id = weapon_id_input;

    IF alpha_stock > bravo_stock THEN
        UPDATE base_inventory
        SET quantity = quantity - quantity_input
        WHERE base_id = 1 AND weapon_id = weapon_id_input;

        UPDATE base_inventory
        SET quantity = quantity + quantity_input
        WHERE base_id = 2 AND weapon_id = weapon_id_input;
    ELSE
        UPDATE base_inventory
        SET quantity = quantity - quantity_input
        WHERE base_id = 2 AND weapon_id = weapon_id_input;

        UPDATE base_inventory
        SET quantity = quantity + quantity_input
        WHERE base_id = 1 AND weapon_id = weapon_id_input;
    END IF;
END//
```

DELIMITER ;

```
✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0040 seconds.)

CREATE PROCEDURE redistribute_stock( IN weapon_id_input INT, IN quantity_input INT ) BEGIN DECLARE alpha_stock INT; DECLARE
bravo_stock INT; -- Get current stock at both bases SELECT quantity INTO alpha_stock FROM base_inventory WHERE base_id = 1 AND
weapon_id = weapon_id_input; SELECT quantity INTO bravo_stock FROM base_inventory WHERE base_id = 2 AND weapon_id = weapon_id_input;
-- Redistribute from surplus to shortage IF alpha_stock > bravo_stock THEN -- Transfer from Alpha to Bravo UPDATE base_inventory SET
quantity = quantity - quantity_input WHERE base_id = 1 AND weapon_id = weapon_id_input; UPDATE base_inventory SET quantity = quantity
+ quantity_input WHERE base_id = 2 AND weapon_id = weapon_id_input; ELSE -- Transfer from Bravo to Alpha UPDATE base_inventory SET
[ Edit ]
```

CALL redistribute_stock(1, 15);
SELECT * FROM base_inventory;

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0056 seconds.)

```
CALL redistribute_stock(1, 15);
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 3 (4 total, Query took 0.0002 seconds.)

```
SELECT * FROM base_inventory;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				base_id	weapon_id	quantity
☐	 Edit	 Copy	 Delete	1	1	45
☐	 Edit	 Copy	 Delete	1	2	20
☐	 Edit	 Copy	 Delete	2	1	65
☐	 Edit	 Copy	 Delete	2	3	5

↑ ☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

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Question 04

```
DELIMITER //

CREATE PROCEDURE distribute_procurement(
    IN weapon_id_input INT,
    IN total_quantity INT
)
BEGIN
    DECLARE remaining INT;
    DECLARE alpha_stock INT;
    DECLARE bravo_stock INT;

    SET remaining = total_quantity;

    WHILE remaining > 0 DO
        SELECT quantity INTO alpha_stock
        FROM base_inventory
        WHERE base_id = 1 AND weapon_id = weapon_id_input;

        SELECT quantity INTO bravo_stock
        FROM base_inventory
        WHERE base_id = 2 AND weapon_id = weapon_id_input;

        IF alpha_stock <= bravo_stock THEN
            UPDATE base_inventory
            SET quantity = quantity + 5
            WHERE base_id = 1 AND weapon_id = weapon_id_input;
        ELSE
            UPDATE base_inventory
            SET quantity = quantity + 5
            WHERE base_id = 2 AND weapon_id = weapon_id_input;
        END IF;

        SET remaining = remaining - 5;
    END WHILE;
END//

DELIMITER ;
```


✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

```
CREATE PROCEDURE distribute_procurement( IN weapon_id_input INT, IN total_quantity INT ) BEGIN DECLARE remaining INT; DECLARE alpha_stock INT; DECLARE bravo_stock INT; SET remaining = total_quantity; WHILE remaining > 0 DO -- Get current stock levels SELECT quantity INTO alpha_stock FROM base_inventory WHERE base_id = 1 AND weapon_id = weapon_id_input; SELECT quantity INTO bravo_stock FROM base_inventory WHERE base_id = 2 AND weapon_id = weapon_id_input; -- Add 5 to base with lowest stock IF alpha_stock <= bravo_stock THEN UPDATE base_inventory SET quantity = quantity + 5 WHERE base_id = 1 AND weapon_id = weapon_id_input; ELSE UPDATE base_inventory SET quantity = quantity + 5 WHERE base_id = 2 AND weapon_id = weapon_id_input; END IF; [...]
```

[Edit]

CALL distribute_procurement(1, 20);
SELECT * FROM base_inventory;

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0081 seconds.)

```
CALL distribute_procurement(1, 20);
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 3 (4 total, Query took 0.0002 seconds.)

```
SELECT * FROM base_inventory;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			base_id	weapon_id	quantity
☐	Edit	Copy	Delete	1	1 65
☐	Edit	Copy	Delete	1	2 20
☐	Edit	Copy	Delete	2	1 65
☐	Edit	Copy	Delete	2	3 5

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

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QUESTION 05

```
DELIMITER //

CREATE PROCEDURE get_weapon_status(
    IN weapon_id_input INT,
    OUT total_stock INT,
    OUT status_label VARCHAR(20)
)
BEGIN
    DECLARE min_stock INT;
    DECLARE base1_stock INT;
    DECLARE base2_stock INT;

    SELECT SUM(quantity) INTO total_stock
    FROM base_inventory
    WHERE weapon_id = weapon_id_input;

    SELECT min_required_stock INTO min_stock
    FROM weapons
    WHERE weapon_id = weapon_id_input;

    SELECT quantity INTO base1_stock
    FROM base_inventory
    WHERE base_id = 1 AND weapon_id = weapon_id_input;

    SELECT quantity INTO base2_stock
    FROM base_inventory
    WHERE base_id = 2 AND weapon_id = weapon_id_input;

    IF base1_stock < min_stock OR base2_stock < min_stock THEN
        SET status_label = 'CRITICAL';
    ELSE
        SET status_label = 'STABLE';
    END IF;
END//

DELIMITER ;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0052 seconds.)

```
CREATE PROCEDURE get_weapon_status( IN weapon_id_input INT, OUT total_stock INT, OUT status_label VARCHAR(20) ) BEGIN DECLARE
min_stock INT; DECLARE base1_stock INT; DECLARE base2_stock INT; -- Get total stock across all bases SELECT SUM(quantity) INTO
total_stock FROM base_inventory WHERE weapon_id = weapon_id_input; -- Get minimum required stock SELECT min_required_stock INTO
min_stock FROM weapons WHERE weapon_id = weapon_id_input; -- Get stock at each base SELECT quantity INTO base1_stock FROM
base_inventory WHERE base_id = 1 AND weapon_id = weapon_id_input; SELECT quantity INTO base2_stock FROM base_inventory WHERE base_id
= 2 AND weapon_id = weapon_id_input; -- Determine status IF base1_stock < min_stock OR base2_stock < min_stock THEN SET status_label
```

[Edit]

```
CALL get_weapon_status(1, @total, @status);  
SELECT @total, @status;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

```
CALL get_weapon_status(1, @total, @status);
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. [?](#)

✓ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT @total, @status;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 [v](#) | Filter rows:

Extra options

@total	@status
130	CRITICAL

☐ Show all | Number of rows: 25 [v](#) | Filter rows:

Query results operations

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Question 06

DELIMITER //

```
CREATE FUNCTION get_risk_level(base_id_input INT)
  RETURNS VARCHAR(20)
  DETERMINISTIC
  BEGIN
    DECLARE low_count INT;

    SELECT COUNT(*) INTO low_count
      FROM base_inventory bi
     JOIN weapons w ON bi.weapon_id = w.weapon_id
    WHERE bi.base_id = base_id_input
    AND bi.quantity < w.min_required_stock;

    IF low_count = 0 THEN
      RETURN 'LOW';
    ELSEIF low_count = 1 THEN
      RETURN 'MEDIUM';
    ELSE
      RETURN 'HIGH RISK';
    END IF;
  END//

DELIMITER ;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0053 seconds.)

```
CREATE FUNCTION get_risk_level(base_id_input INT) RETURNS VARCHAR(20) DETERMINISTIC BEGIN DECLARE low_count INT; -- Count weapons below
minimum required stock SELECT COUNT(*) INTO low_count FROM base_inventory bi JOIN weapons w ON bi.weapon_id = w.weapon_id WHERE bi.base_id
= base_id_input AND bi.quantity < w.min_required_stock; -- Determine risk level IF low_count = 0 THEN RETURN 'LOW'; ELSEIF low_count = 1
THEN RETURN 'MEDIUM'; ELSE RETURN 'HIGH RISK'; END IF; END;
```

[Edit inline] [Edit] [Create PHP code]

```
SELECT get_risk_level(1);
SELECT get_risk_level(2);
```

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✓ Showing rows 0 - 0 (1 total, Query took 0.0007 seconds.)

`SELECT get_risk_level(1);`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Extra options

get_risk_level(1)
HIGH RISK

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Query results operations

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⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✓ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

`SELECT get_risk_level(2);`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Extra options

get_risk_level(2)
HIGH RISK

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Query results operations

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QUESTION 07

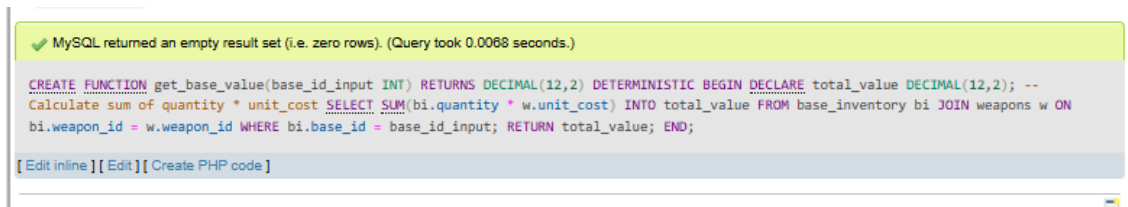
DELIMITER //

```
CREATE FUNCTION get_base_value(base_id_input INT)
  RETURNS DECIMAL(12,2)
  DETERMINISTIC
  BEGIN
    DECLARE total_value DECIMAL(12,2);

    SELECT SUM(bi.quantity * w.unit_cost) INTO total_value
      FROM base_inventory bi
     JOIN weapons w ON bi.weapon_id = w.weapon_id
    WHERE bi.base_id = base_id_input;

    RETURN total_value;
  END//

DELIMITER ;
```

A screenshot of a MySQL query execution result in a web interface. The top bar is green and contains a checkmark icon and the text "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0068 seconds.)". Below this, the SQL code is displayed in a light blue box with syntax highlighting. The code is the same as the one in the previous block. At the bottom of the code box, there are three links: "[Edit inline]", "[Edit]", and "[Create PHP code]".

```
✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0068 seconds.)

CREATE FUNCTION get_base_value(base_id_input INT) RETURNS DECIMAL(12,2) DETERMINISTIC BEGIN DECLARE total_value DECIMAL(12,2); --
Calculate sum of quantity * unit_cost SELECT SUM(bi.quantity * w.unit_cost) INTO total_value FROM base_inventory bi JOIN weapons w ON
bi.weapon_id = w.weapon_id WHERE bi.base_id = base_id_input; RETURN total_value; END;

[ Edit inline ] [ Edit ] [ Create PHP code ]
```

```
SELECT get_base_value(1);
SELECT get_base_value(2);
```

⚠️ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✔️ Showing rows 0 - 0 (1 total. Query took 0.0008 seconds.)

`SELECT get_base_value(1);`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

`get_base_value(1)`
178000.00

☐ Show all | Number of rows: 25 | Filter rows:

Query results operations

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⚠️ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✔️ Showing rows 0 - 0 (1 total. Query took 0.0005 seconds.)

`SELECT get_base_value(2);`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

`get_base_value(2)`
203000.00

☐ Show all | Number of rows: 25 | Filter rows:

Query results operations

☐ Show all | Number of rows: 25 | Filter rows:

Query results operations

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